

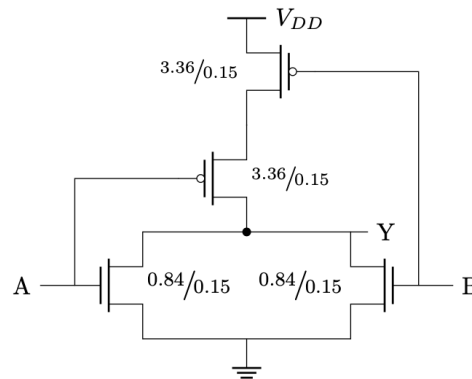
Experiment 7: NOR gate – schematic and layout extracted simulation

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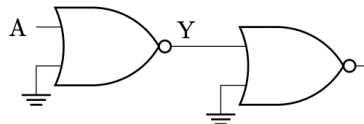
Assignment Questions

EE5311 Digital IC Design: Assignment 7 NOR gate – schematic and layout extracted simulation

1. Construct a two-input NOR gate shown below. Use appropriate number of fingers for all the devices considering the layout constraints.



The NOR gate drives another NOR gate of same size as shown below:



The input A is a pulse between 0 and V_{DD} , with a rise and fall time of 5 ps and a pulse width of 400 ps. Input B is connected to ground.

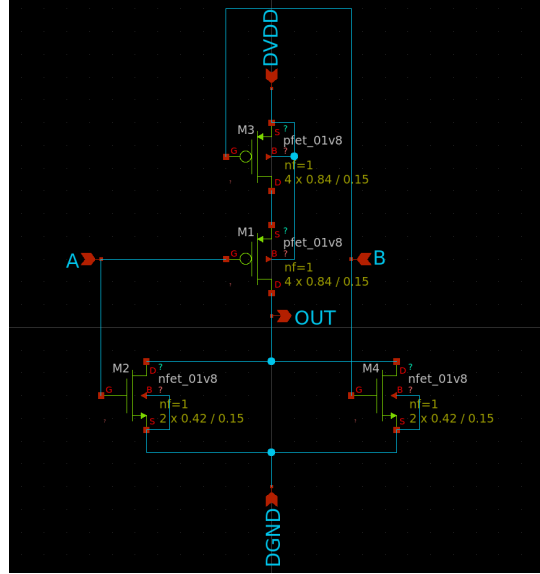
- (a) Obtain the delay of the NOR gate for $V_{DD} = 1.8\text{V}$. How does this delay compare against the one calculated using the logical effort expression?
- (b) Draw the layout containing two NOR gates connected as shown above and measure the layout extracted delay for $V_{DD} = 1.8\text{V}$.

Figure 1: Assignment Questions for Experiment 7

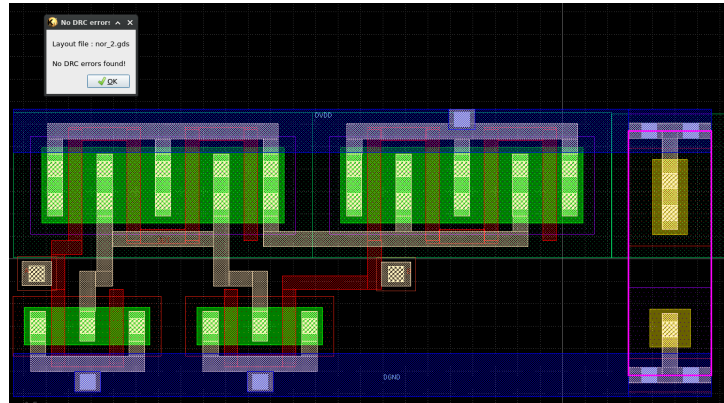
Q1(a). NOR Gate Layout and Delay Measurement

Q1(a.1) Schematic, DRC and LVS Clean Layout

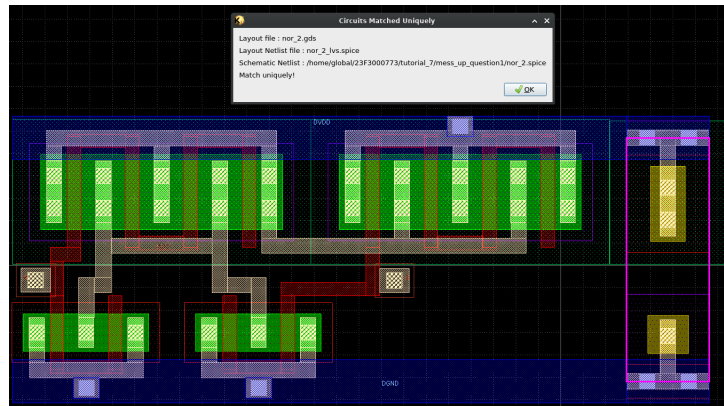
The NOR gate uses $(W/L)_p = 3.36/0.15 \text{ } \mu\text{m}$ and $(W/L)_n = 0.84/0.15 \text{ } \mu\text{m}$ devices. The schematic, DRC clean, and LVS clean screenshots are shown below.



(a) NOR gate schematic.



(b) DRC clean layout.



(c) LVS clean layout.

Figure 2: Schematic, DRC and LVS clean layout of the NOR gate.

Q1(a.2) Delay Measurement with Parasitics

The delay was measured using an extracted netlist, with 50% transition points of V_{in} and V_{out} .

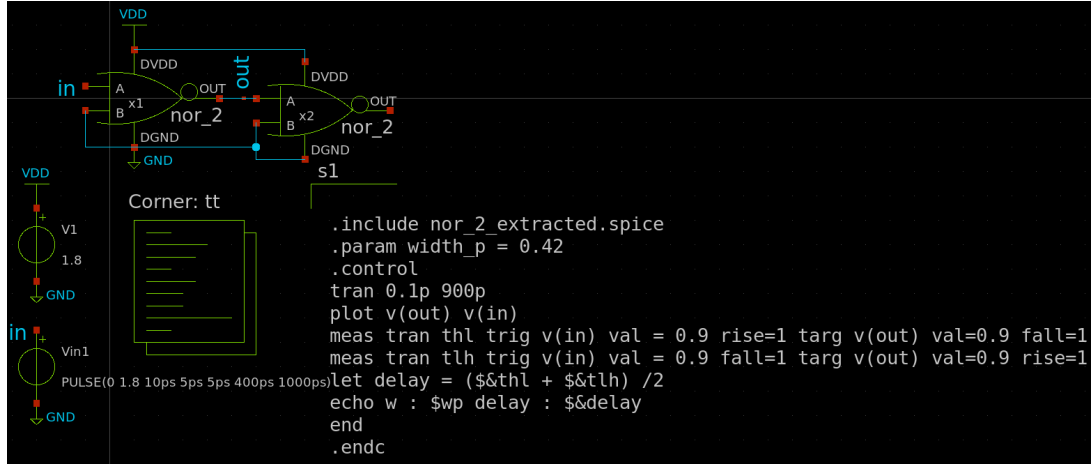


Figure 3: Delay measurement testbench for NOR gate.

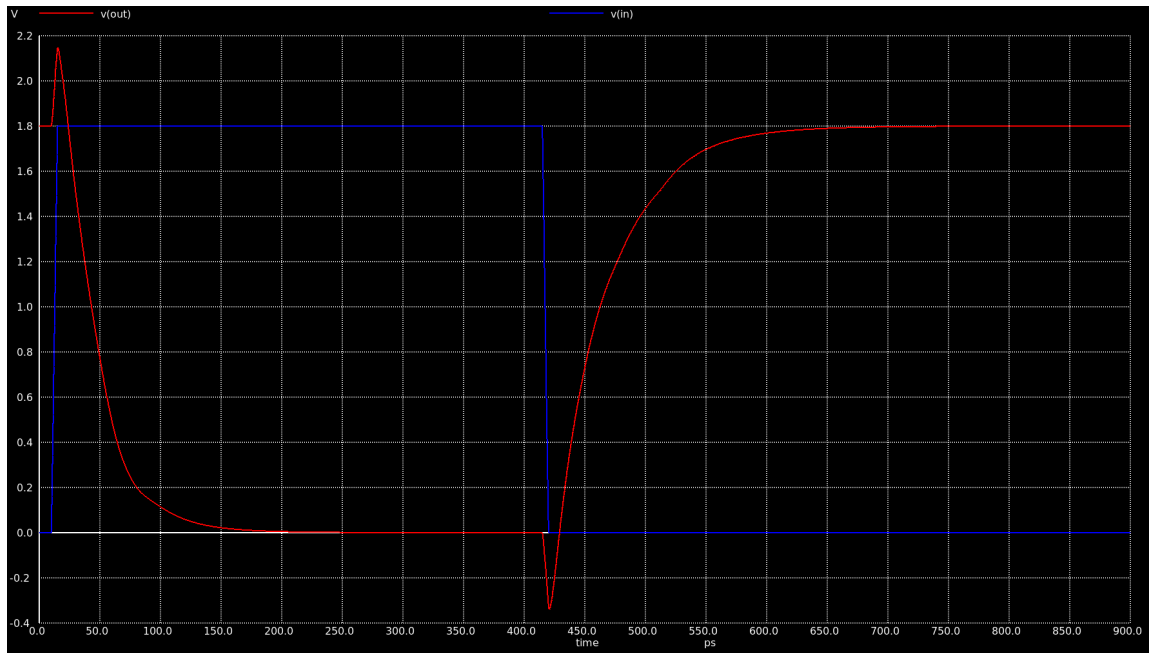


Figure 4: Transient simulation showing V_{in} and V_{out} for delay extraction.

Table 1: Post-Layout NOR Gate Propagation Delay

Parameter	Pre-Layout (ps)	Post-Layout (ps)
t_{pLH}	36.34	39.73
t_{pHL}	30.33	33.55
t_{pd} (avg.)	33.33	36.64

Q1(a.3) Comparing measured delay against calculated delay.

delay from (A to out) $\Rightarrow (gh+P)T$

Delay calculation using logical effort (g)

Calculation of delay of inverter;

$$T_{inv} = \frac{1}{2} (t_{phl} + t_{plh})$$

$$I_{DSATn} = \frac{\mu_{ncox} \eta}{2} \frac{W_n}{L} \frac{(V_{DD} - V_{th})^2 E_{chn}}{V_{DD} - V_{th} + E_{chn}}$$

$$I_{DSATn} = 1.645 \times 10^{-4} A$$

$$I_{DSATp} = \frac{\mu_{ncox} W_p}{2} \frac{V_{DD}}{2} (1.0)^2 \left(\frac{6.6 \times 0.15}{0.1 + 0.66 \times 0.15} \right)$$

$$\Rightarrow I_{DSATp} = 1.183 \times 10^{-4} A$$

$$T_{inv} = \frac{C_{VDD}}{4} \left(\frac{1}{I_{DSATp}} + \frac{1}{I_{DSATn}} \right)$$

$$\Rightarrow T_{inv} = 9.8 ps$$

$$\Rightarrow \text{delay of NOR gate} = (gh+P)T \quad \left. \begin{array}{l} g = \frac{5}{3}; h=1; P \Rightarrow 2 \end{array} \right\}$$

$$\Rightarrow \text{delay of NOR gate} = \left(\frac{5}{3} + 2 \right) \times 9.8 ps$$

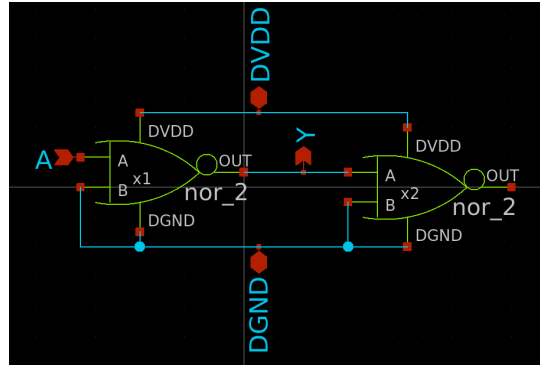
$$\text{delay}_{NOR} = 36.03 ps$$

Figure 5: Delay calculation of the NOR gate

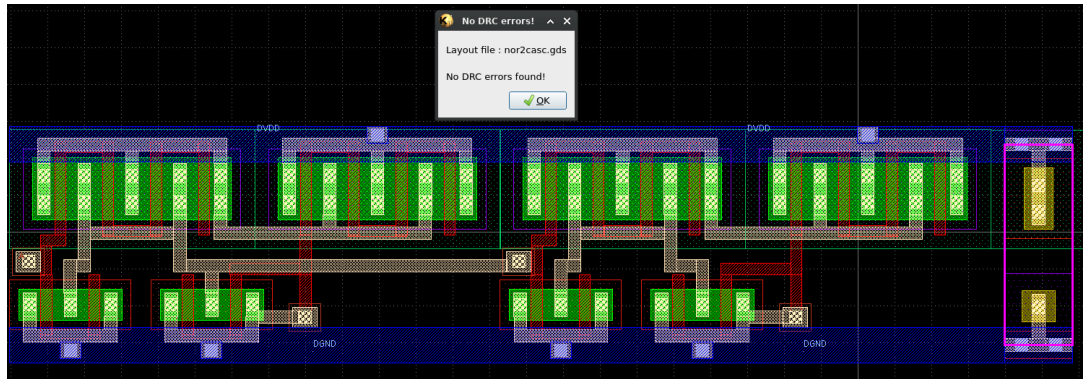
Q1(b). Two Cascaded NOR Gates Layout and Delay Measurement

Q1(b.1) Schematic, DRC and LVS Clean Layout

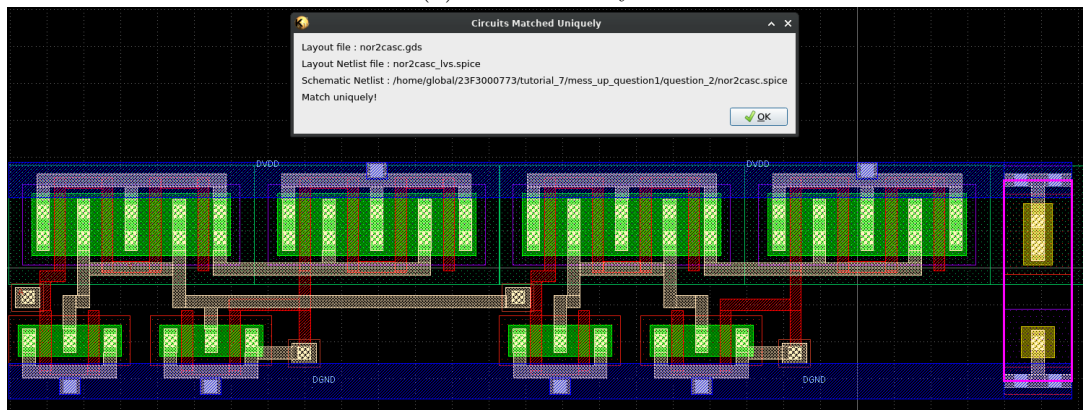
The layout for two cascaded NOR gates of the same size was designed. The schematic, DRC clean, and LVS clean screenshots are shown below.



(a) Cascaded NOR schematic.



(b) DRC clean layout.



(c) LVS clean layout.

Figure 6: Schematic, DRC and LVS clean layout of cascaded NOR gates.

Q1(b.2) Delay Measurement with Parasitics

The propagation delay of the cascaded NOR gates was measured from extracted netlists.

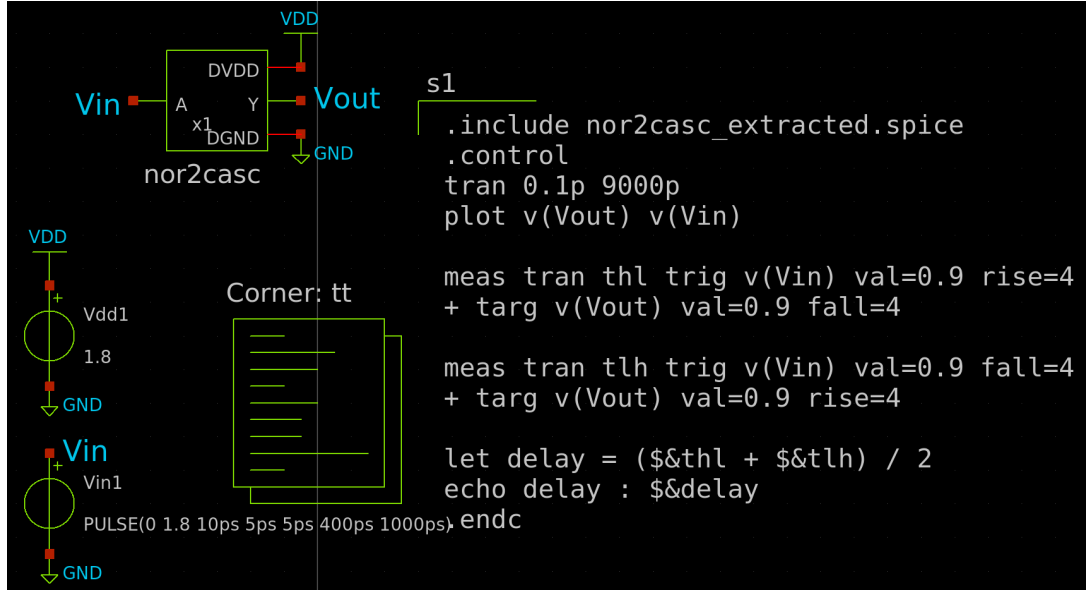


Figure 7: Delay measurement testbench for cascaded NOR gates.

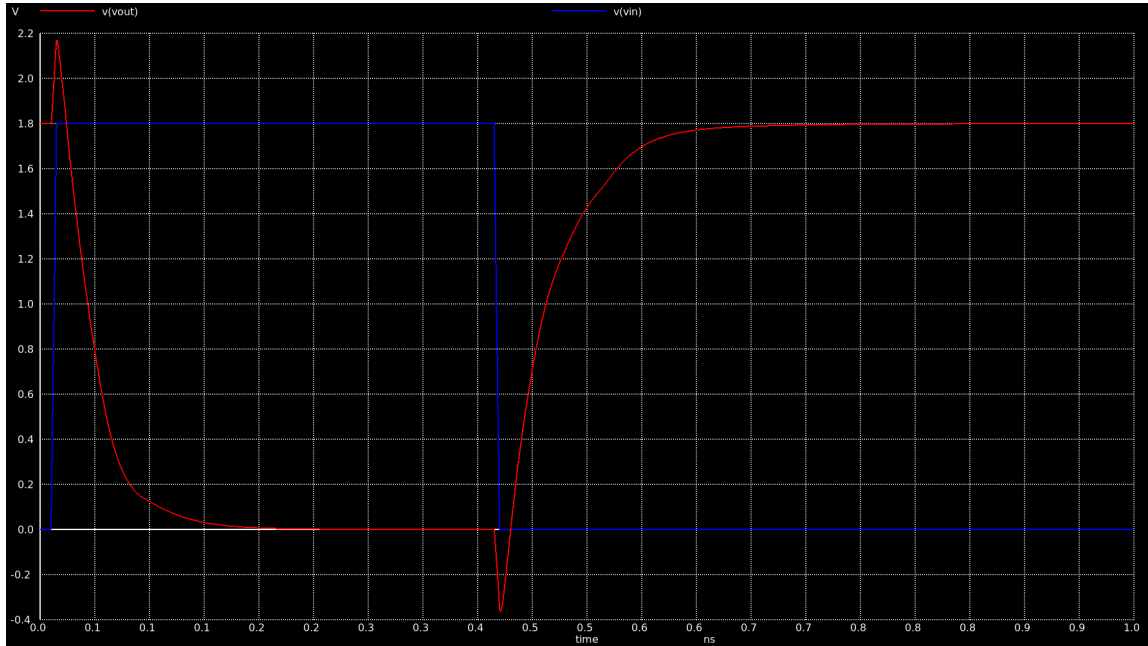


Figure 8: Transient simulation showing V_{in} and V_{out} for cascaded NOR gates.

Table 2: Post-Layout Cascaded NOR Gate Propagation Delay

Parameter	Pre-Layout (ps)	Post-Layout (ps)
t_{pLH}	36.34	40.09
t_{pHL}	30.33	34.07
t_{pd} (avg.)	33.33	37.08

Conclusion

Post-layout simulations showed an increase in NOR gate propagation delay compared to pre-layout results, mainly due to parasitic capacitances and resistances.